

Natural Language Processing

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Steps for NLP

1. Text Collection
2. Text Preprocessing
 - a. Converting everything to lowercase
 - b. Remove punctuations (comma, fullstops etc.)
 - c. Stop words removal (is, am, and etc.)
 - d. Tokenization (dividing whole corpus into individual tokens)
 - e. Lemmetization and Stemming (converts the words(tokens) into its base form (eg. learning --> learn))
3. Feature Extraction
 - a. Vectorization (converting the tokens into vectors. (Eg. Word2Vec,)
Or
 - b. Embeddings (Embedding/ Mapping the individual words to its numeric representation.)
4. Modeling

Natural Language Processing [NLP] → NLP is a branch of Artificial Intelligence (AI) that empowers computers to comprehend, interpret, and generate human language. It bridges the gap between human communication and machine understanding, powering essential tools like search engine, spam filtering and predictive text.

How NLP works [The 5-step process] -

NLP breaks down language into discrete components to extract meaning and intent using five key phases:-

- ① Lexical Analysis → Break down the text into paragraphs, sentences and individual words [tokens], analyzing the structure and form of each word.
- ② Syntactic Analysis → Assess sentence grammar, ensuring words are arranged in a way that aligns with formal language rules.
- ③ Semantic Analysis → Extracts the literal meaning of words and maps them to their

of the text and maps individual words to actual objects and actions.

④ Discourse Integration → Looks at how a sentence's meaning changes depending on the sentence that comes before or after it.

⑤ Pragmatic Analysis → Interprets what the author actually means beyond the literal definition, considering context, sarcasm and real-world knowledge.

Two Core Components

NLP encompasses two primary, interconnected disciplines:-

① Natural Language Understanding [NLU] → Focus on reading comprehension. NLU maps human inputs into a structured framework so the machine can determine intent and specific meaning.

② Natural Language Generation [NLG] → Focuses on "writing". NLG takes structured computer data and translates it into fluid human-readable text.

Key NLP tasks -

Machines are trained to perform several highly specialized

tasks to process the language:-

- ① Sentiment Analysis → Determines whether a piece of text is positive, negative or neutral. [mainly used for tracking social media reviews]
- ② Named Entity Recognition (NER) → Identifies specific people, places, dates and organisations within a text.
- ③ Machine Translation → Translates text or speech from one language to another [eg. Google Translate]
- ④ Text Summarization → Condenses long articles, books or documents into short, readable overview.